Joint Contribution of Multiple Equatorial Waves to Tropical Cyclogenesis over the Western North Pacific

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ABSTRACT

A composite study is performed to examine the differences in equatorial wave behaviors and large-scale background patterns during tropical cyclone (TC) genesis. After removing TC contamination, Madden–Julian oscillation (MJO), equatorial Rossby (ER) wave, mixed Rossby–gravity (MRG) wave, and tropical depression (TD)-type disturbance (jointly referred to as the MT wave) are quantified to evaluate the attribution of TC geneses. Given that TC geneses are attributed to a single wave or multiple waves, the eight categories are specified based on the moderate thresholds. The TC geneses related to multiple waves are roughly twice as many as those related to a single wave. The MT wave alone accounts for a minor proportion of TC geneses without collaboration with other larger-scale waves. The mean TC genesis location related to ER wave shifts to higher latitudes, and the TC geneses attributed to both of MJO and MT waves are more concentrated at the west. The single-wave categories are characterized by a zonally propagating component with a large spatial scale. In contrast, the joint contribution from more than one wave type favors creating a coherent environment with enhanced low-level cyclonic vorticity, horizontal convergence, or vertical easterly shear in a preferred region. Consequently, the waves have a more robust structure and a more northwestward-propagating component. Correspondingly, the TC geneses in the MJO–ER category tend to occur within a monsoon trough dominated by cyclonic circulation. For the MJO–MT category, the background field exhibits a confluence pattern with a monsoon trough to the west and easterly flows to the east. The collaboration of the ER and MT waves facilitates TC geneses within an easterly environment in the southern flank of the subtropical high.

1. Introduction

In the tropics, it is commonly observed that tropical cyclone (TC) genesis is closely related to multiscale equatorial waves. From intraseasonal to synoptic scale, the family of equatorial and near-equatorial waves could not only provide a favorable environment in which mesoscale deep convection systems tend to be well organized into a self-sustaining vortex system, but also feed initial disturbance seedlings as TC genesis precursors. Some studies have been designed to provide probabilistic forecasts of tropical cyclone, which develops in association with convectively coupled equatorial waves and intraseasonal oscillations by monitoring the modes of coherent synoptic to intraseasonal tropical variability (e.g., Wheeler and Hendon 2004; Roundy and Schreck 2009).

Growing evidence reveals that a large fraction of TC geneses over the western North Pacific (WNP) are significantly modulated by tropical oscillations or waves. Among these modes, Madden–Julian oscillation (MJO; Madden and Julian 1994) is the dominant mode of intraseasonal convective variability in the tropics. It features a planetary-scale disturbance in both convection and circulation (predominantly zonal wind), which has characteristics of eastward propagation along the equator across the Indian Ocean and the western Pacific Ocean with a period of 30–60 days. MJO can play an important role in modulating TC activity over many ocean basins,

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such as the Indian Ocean, the WNP, the eastern North Pacific, and the Caribbean Sea (e.g., Bessafi and Wheeler 2006; Liebmann et al. 1994; Maloney and Hartmann 2000). The MJO westerly phase could modulate environmental fields by producing cyclonic horizontal shear of low-level zonal wind and increasing midlevel moisture, resulting in statistically significantly more TCs in the active phase than in the inactive phase of MJO. Other than the direct modulation on TC genesis, MJO could also act as a key ingredient for establishing a convective envelope, in which higher-frequency waves experience amplification by scale contraction and energy accumulation, favorable for TC geneses (Nakazawa 1986). From the energetic perspective, Maloney and Dickinson (2003) found that the perturbation kinetic energy corresponding to tropical depression (TD)-type disturbance is significantly higher during a MJO westerly than easterly period. Meanwhile, the MJO westerly phase can enhance barotropic conversion from mean kinetic energy to perturbation kinetic energy and baroclinic conversion from perturbation available potential energy to perturbation kinetic energy, which helps explain MJO modulation on TCs. In an idealized numerical model, Aiyyer and Molinari (2003) also depicted that the mixed Rossby–gravity (MRG) waves can transit to the TD-type disturbances and grow rapidly when the MRG waves, which propagate westward along the equator, encounter the tropical asymmetric basic state initiated by the MJO.

In addition to the MJO mode in the tropics, equatorial Rossby (ER) waves also carry substantial variance in boreal summer and fall, which could contribute to the origin of quasi-biweekly oscillations (Chen and Sui 2010). Following the spectral analyses, Roundy and Frank (2004) found ER-filtered convection represented by outgoing longwave radiation and precipitable water can stand out clearly in the wavenumber–frequency domain, especially in the Northern Hemisphere. The ER spectral region appears to be dominated by waves of meridional mode $n = 1$ with an antisymmetric structure about the equator. After examining the relationship between tropical wave activity and TC geneses over major ocean basins, Frank and Roundy (2006) found that, over the WNP, the coherent annual cycle in ER wave activity is coincident well with the TC season. In a case study, Molinari et al. (2007) documented a packet of ER waves that lasted 2.5 months with a period of 22 days, a wavelength of 3600 km, and a westward phase speed of $1.9\,\mathrm{m\,s^{-1}}$. At least 8 of the 13 WNP TCs that formed during this period are associated with the ER packet by enhancing the background convection and low-level cyclonic vorticity. Schreck and Molinari (2009) also found that three sets of twin TCs developed in the central Pacific within a single month. The corresponding equatorial westerlies and convective anomalies were modulated by convectively coupled $n = 1$ ER waves.

Another prominent tropical wave mode related closely to TC genesis is referred to as a TD-type disturbance, which is characterized by a northwestward propagation with a wavelength of approximately 3000 km and a 3–8-day time scale. The origin of the TD-type disturbance may arise from off-equatorial westward-propagating easterly waves (Reed and Recker 1971; Reed et al. 1977), Rossby wave energy dispersion of a preexisting mature TC (Li and Fu 2006; Li et al. 2006), and a synoptic wave train transiting from equatorially trapped MRG waves (Takayabu and Nitta 1993; Chen and Huang 2009). The TD-type disturbances with different origins are all likely to contribute to TC geneses. In contrast to TC geneses over the WNP, TCs in the North Atlantic are commonly induced by easterly waves. Landsea (1993) documented approximately 60% of tropical storms, 60% of weak hurricanes, and 80% of intense hurricanes in the Atlantic evolved from African easterly waves. Fu et al. (2007) examined TC genesis events over the WNP during the 2000 and 2001 typhoon seasons, and found that, among 34 cyclogenesis cases, 6 are associated with the energy dispersion associated with a preexisting TC. Through the synoptic analysis in a case study, Dickinson and Molinari (2002) attributed several TC geneses over the WNP to the wave transition from the equatorial MRG waves. Under the environment related to the monsoon trough, the westward-propagating MRG waves moved away from the equator and gradually transformed to the off-equatorial disturbances. Consequently, 3 TCs formed in the cyclonic gyres of an off-equatorial synoptic wave train. Overall, off-equatorial TD-type disturbances account for a considerable percentage of TC geneses by providing plenty of seedlings of initial disturbances.

In contrast to the above-mentioned waves, the linkage between TC geneses and Kelvin waves over the WNP is less clear, and the corresponding classic Kelvin wave structure is hard to discern by a Kelvin wave band composite relative to TC genesis locations (Frank and Roundy 2006). Moreover, Kelvin waves over the WNP exhibit no significant annual cycle, without a good phase relationship with TC season. However, this does not necessarily mean that Kelvin waves do not affect TC genesis. Some cases can still be detected close to the equator, especially when Kelvin waves are embedded within a MJO convective envelope (e.g., Schreck and Molinari 2011). However, these events do not occur frequently enough to produce a strong signal in the statistics.

In summary, a great deal of research has revealed that TC geneses over the WNP can be mainly attributed to MJO, ER waves, TD-type disturbances, and MRG waves. For brevity, these modes with the range of intraseasonal

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to synoptic scales are referred to as equatorial waves because they reside near equatorial regions and share wavelike properties. Most of previous studies focused on the relationship between individual wave type and TC geneses or case analysis of TC geneses affected by multiple waves. In reality, it is common that multiple wave types coexist and exert a joint influence on TC geneses over the WNP (e.g., Schreck et al. 2011; Schreck and Molinari 2011). From a statistical perspective, the extent to which multiple wave types can contribute to TC geneses, and the differences in wave behaviors for a single wave and multiple waves contributing to TC geneses are not well understood.

In addition, another intriguing question arises as to whether the large-scale circulation patterns that are associated with multiple wave types also exhibit clear differences during TC genesis processes. Ritchie and Holland (1999) classified the large-scale environmental flow patterns preferred for TC geneses into five categories: monsoon shear line, monsoon confluence region, monsoon gyre, easterly wave, and Rossby energy dispersion. On one hand, since large-scale background circulation can play a crucial role in modulating equatorial wave characteristics, many wave types tend to persist within favorable large-scale circulation patterns. On the other hand, the strong wave-triggered convection and its associated dynamic structure over long and large enough temporal and spatial scales can also alter large-scale circulation patterns. It suggests that the interaction of equatorial waves and background circulation can produce a favorable background to enhance the potential for TC geneses. The understanding of the correspondence between equatorial waves and large-scale circulation patterns during different types of TC genesis processes can further improve our knowledge of the tropical atmospheric conditions that are favorable for TC geneses and help enhance the predictability of TCs on the synoptic and intraseasonal scales.

In section 2, the data and analysis method are introduced. TC geneses attributed to the different wave types are categorized, and TC genesis statistics in the each category are presented in section 3. The wave features related to TC geneses in different categories are compared in section 4. Section 5 explores the large-scale environmental circulation in the individual category prior to TC geneses, making an attempt to shed light on the linkage of equatorial waves and background circulations that are favorable for TC geneses. Conclusions are given in section 6.

2. Data and methodology

As the dataset and procedure used in Schreck et al. (2011), the Tropical Rainfall Measuring Mission (TRMM) Multisatellite Precipitation Analysis (TRMM 3B42; Huffman et al. 2007) is employed as a proxy of convection. To be compatible with TC data and improve computational efficiency, the data are interpolated into 6-hourly 1° latitude–longitude grids that can well resolve tropical waves associated with TC geneses. The atmospheric circulation is represented by the Interim European Centre for Medium-Range Weather Forecasts (ECMWF) Re-Analysis (ERA-Interim). The 6-hourly ERA-Interim products produced by an improved atmospheric model and an assimilation system with a horizontal resolution of 1.5° (http://data-portal.ecmwf.int/data/d/interim_daily/levtype=pl/) are used here. The ERA-Interim data are also interpolated to 1° to keep the consistency with TRMM data.

TC information is obtained from the National Climatic Data Center's (NCDC's) International Best Track Archive for Climate Stewardship (IBTrACS version 3, release 4), which incorporates TC data from operational centers around the world. The time, at which the disturbance with maximum 10-min sustained winds of $13\,\mathrm{m\,s^{-1}}$ was first classified as a tropical depression, is defined as the time of TC genesis. At that time, the record of TC locations is identified as TC genesis locations. Considering the seasonality of tropical waves and TC activity (Frank and Roundy 2006), the present study will focus on the TC geneses over the WNP during the warm seasons from May to November during the years of 1998–2010. Based on the statement of Schreck et al. (2011), only TC geneses in the region covering from the equator to 20°N and 120°E to the date line are considered to exclude those less related to equatorial waves. By selection, there are a total of 214 TCs from May to November during the period of 13 years.

As an extreme event, TC could significantly modulate the ensemble mean and variability, and contaminate the character of tropical waves to produce artificial signals in the TC-prone region, especially during the TC peak seasons (e.g., Hsu et al. 2008; Schreck et al. 2011). Therefore, before investigating the relationship between equatorial waves and TC geneses, the contamination from TCs should be diminished by removing the TC-related anomalies. Following the method of Schreck et al. (2011), first, the rainfall anomalies were obtained by subtracting the climatological rainfall from the total rainfall. And then, a Gaussian weighting function was multiplied to the rainfall anomalies centered in each TC genesis position:

$$w(x, y) = 1 - 2^{-[r(x,y)/R]^2},$$

where $r(x, y)$ is the distance between a given grid point and the TC center, and R is the radius at half maximum of the weighting function. Because the details of the

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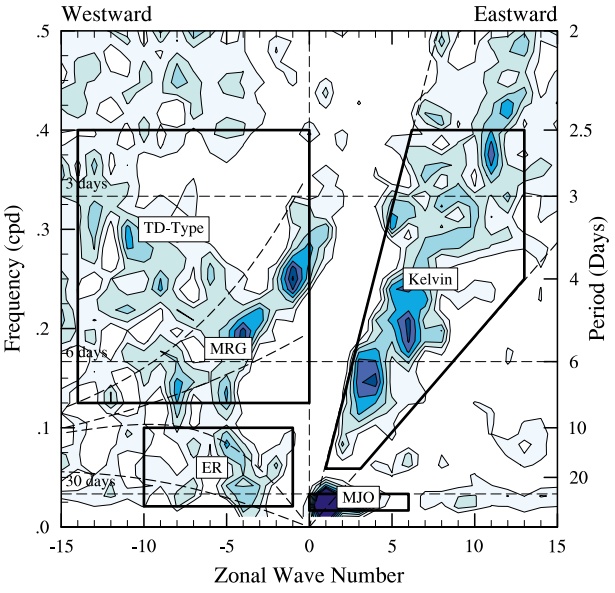


FIG. 1. Normalized wavenumber–frequency spectrum of TRMM observations averaged between 0° and 20°N from May to November for the period of 1998–2010 by a red noise. The thick black lines encompass the regimes of filter bands. The thin dashed lines denote shallow-water dispersion curves with equivalent depths of 8 and 90 m.

removal cannot impact significantly the attribution results, $R = 500$ km is used in this study. In such a way, all of the TC-related rainfall above climatology is removed in the TC center and the removal fraction decreases with radius. The TC-removal background rainfall can be obtained by adding the TC-removal anomalies to the climatology.

Similar to the difference between the total spectra of the original data and the TC-removed data in the study of Schreck et al. (2011, see their Fig. 6), the pronounced percentage contribution from TC-related signals falls within the westward-propagating synoptic wave bands (not shown). To specify the filtering domain, Fig. 1 shows the normalized wavenumber–frequency spectrum of the TC-removed rainfall anomalies averaged between 0° and 20°N by a red noise. The boxes in the figure outline the filter bands used in this study, which capture most of spectrum power that stand out above the red background. In this study, the impact of Kelvin waves is ignored in the examination of TC genesis attribution to equatorial waves for the sake of less clear relationship between Kelvin waves and TC geneses during TC peak seasons over the WNP, as previously mentioned. The bands of the MRG wave and the TD-type disturbance have overlapping regimes because of their proximity in terms of frequency and wavenumber, and hence, it is hard to entirely separate these two higher-frequency wave types. For simplicity, the MRG wave and TD-type disturbance are combined into one regime (referred to as MT wave for convenience) for filtering. The MT wave band covers periods from 3 to 8 days and westward propagation at wavenumbers of −14 to 0. The ER band has periods of 10–48 days and wavenumbers of −10 to −1, and the MJO band for periods of 30–60 days and eastward propagation at wavenumbers of 0–6. Although these filters are slightly different from those in precious studies (e.g., Frank and Roundy 2006; Schreck et al. 2011), the present filters can still capture most of the wave variance and obtain qualitatively consistent results. Likewise, the wind fields are also filtered through the same wavenumber–frequency domains as those employed to the rainfall anomalies. However, it should be noted that the wave signals passing through the filter are not necessarily representative of these equatorial waves. They could include any disturbances within this spatial and temporal domain. Nevertheless, based on the attribution thresholds corresponding to the wave types described in the next section, TC geneses over the WNP can be classified into eight categories, in which the different combination of wave types acts as a dominant precursor of TC geneses.

The statistical significance was evaluated using a bootstrap test similar to Frank and Roundy (2006). In this test, null cases were identified using the dates of all TC geneses except the samples in the tested category. Null composites were generated by randomly selecting the same number of dates from the null cases as had been in the tested category. This procedure was repeated to calculate 1000 null composites. Using a two-tailed test, the 90% confidence intervals are given by the 50th ranked mean and 95th ranked mean.

3. TC categorization and statistics

The comparison of standard deviation of three major equatorial wave types before and after removing the contamination related to TCs demonstrates that TC contamination is concentrated around the regions where TCs are more densely populated, while the contribution percentages related to TC occurrence differ among these wave types (not shown). In contrast to the other two wave types, the MT wave band has the largest standard deviation along a narrow strip centered at 10°N, whereas TC-related contribution in the MT wave band exceeds 40% of standard deviation near the Philippines, twice as large as that in the MJO band. This indicates that the synoptic wave band is more vulnerable to TC contamination compared to other wave bands with larger spatial and temporal scales. These results are in good agreement with those found in previous studies (cf. Fig. 3 in Schreck et al. 2012).

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Because the standard deviations differ with respect to wave types and geographic position, the threshold value, which depends on both wave types and TC genesis locations, was specified to evaluate the role of equatorial waves in cyclogenesis. One TC genesis is considered to be attributed to one given wave type when the mean wave–filtered anomaly within the 4° × 4° domain centered in TC genesis locations is more than the half standard deviation of the corresponding wave type. One squared domain is required in order to ensure that the major TC circulation, instead of one grid point, is embedded within the wave-filtered background environment. Since the classification is sensitive to the threshold, the half standard deviation of the corresponding wave type is specified as a criterion based on the following considerations. Requiring a positive anomaly (greater than zero standard deviation) alone is too weak to ensure convective enhancement. At the other extreme, the requirement of one standard deviation is so strict that most of TC geneses would not be attributed to any wave precursor.

According to the above criterion, the TC geneses can be classified into eight categories. The three categories, in which TC geneses are only attributed to a single wave type at the time of TC genesis (i.e., the corresponding wave anomaly exceeds half of its standard deviation in the given TC genesis location), are referred to as MJO, ER, and MT categories, respectively. Another three categories consist of TC geneses attributed to any two wave types (i.e., MJO–ER, MJO–MT, and ER–MT categories), while the third wave anomaly is below the threshold value. The remaining two categories are the MJO–ER–MT and None categories, in which all or none of the three wave types contribute to TC geneses, respectively. Based on the classification, all 214 TC geneses can be partitioned into one individual group without overlapping, avoiding cross impacts on the composites discussed later.

The TC genesis percentage in each category is depicted in Fig. 2. The TC geneses without any impact of wave types (the None category) are the fewest among the eight categories, accounting for 5.6%. The contribution from more than one wave type (i.e., the MJO–ER, MJO–MT, ER–MT, and MJO–ER–MT categories) is responsible for 144 TC geneses and 67% of total cases, whereas the number of TC geneses attributed to a single wave type is 58, less than half of the TC geneses that are attributed to multiple wave types. In the categories involving a signal wave type, it seems to be surprising that the attribution related to the ER wave type appears to be the strongest, while the number of TC geneses associated with MJO is comparable to that associated with MT. In previous studies (e.g., Frank and Roundy 2006; Schreck et al. 2012), the MT wave type was regarded as

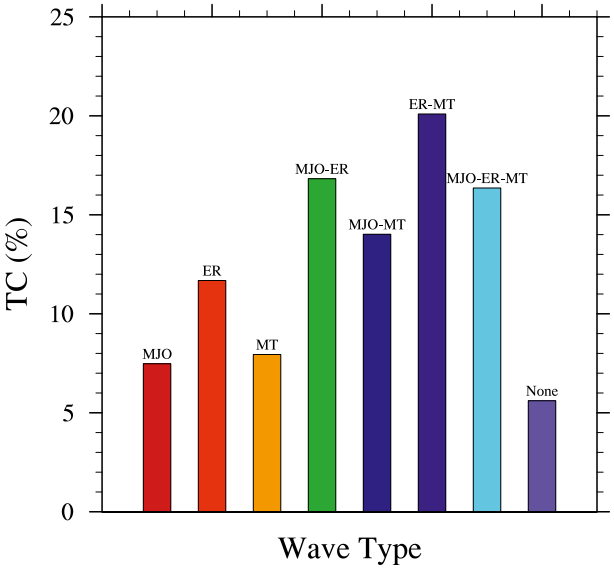


FIG. 2. The percentage of TC geneses in the MJO, ER, MT, MJO–ER, MJO–MT, ER–MT, MJO–ER–MT, and None categories. Note that no TC cases are overlapped within the different categories. Their summation is 100%.

a dominant precursor for TC geneses. The discrepancy, on one hand, lies in that TC geneses related to the MT wave type were not separated into distinct categories with or without the collaboration of other wave types in previous studies. The present study implies that, without the collaboration of other wave types, the MT wave type alone is less related to TC geneses because of the lack of wave amplification. On the other hand, the definition of threshold criterion largely determines the TC genesis percentage in each category. Schreck et al. (2012) employed a constant threshold value for all wave types to assess the role of equatorial waves in TC genesis. In contrast, the threshold values in the present study depend on wave types and TC genesis locations, considering that the standard deviations vary in terms of wave types and geographic distribution. As a result, the mean threshold values in the MJO, ER, and MT categories are 2.1, 3.6, and 4.5 mm day^{-1}, respectively. Therefore, the relatively strict criterion applied in the MT category could partly lead to the decrease in the TC genesis percentage associated with the MT wave type. Besides, TC geneses can be commonly observed within a synoptic environment evolved from monsoon gyre. Chen et al. (2004) documented that the majority (roughly 70%) of TC geneses over the WNP are linked to monsoon gyre modulated by summer monsoon circulation. Given that monsoon gyre has a temporal and spatial pattern close to the ER wave type, the ER wave type could account for a large percentage of TC geneses in this study.

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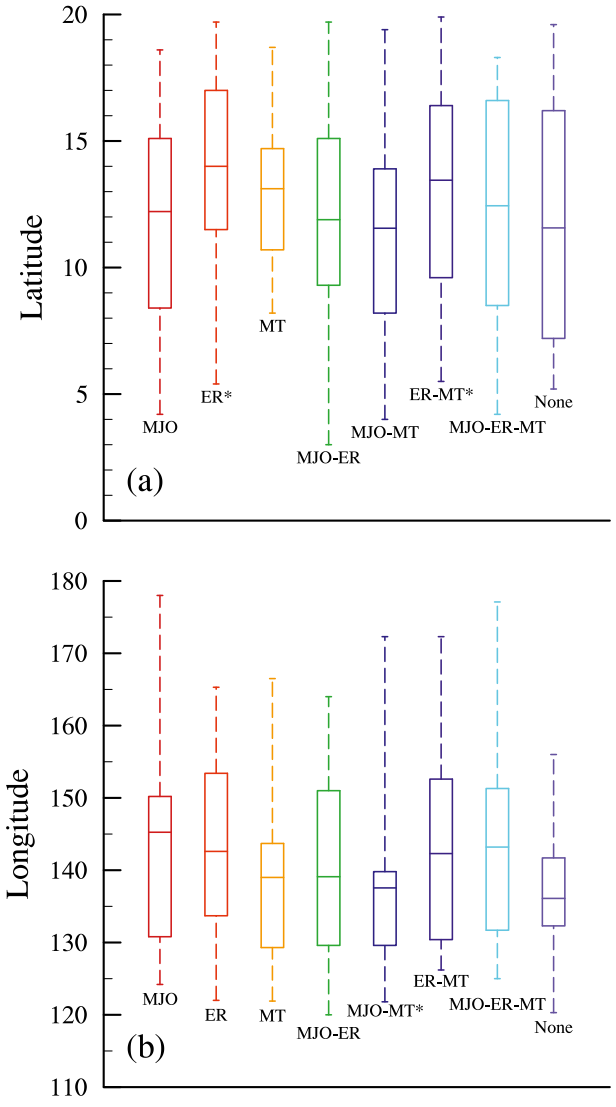


FIG. 3. Boxplots of mean (a) latitude and (b) longitude of the TC geneses in the eight categories. The vertical bar shows the full range of the distributions, and the box shows the interquartile range of 25%–75% of the all cases. The bold horizontal line indicates the mean. The categories with an asterisk represent mean latitude or longitude that deviates significantly from null hypothesis at the 90% confidence level.

Figure 3 exhibits box plots for the distribution range of TC genesis latitudes and longitudes. Using the bootstrap test, most of TC genesis locations have no significant deviation from null hypothesis, except for the mean latitudes in the ER and ER–MT categories and the mean longitude in the MJO–MT category. The mean latitudes in the ER and ER–MT categories are significantly higher compared to those in other categories at 90% confidence level. This may be partly due to the more meridionally extensive structure for $n = 1$ ER waves, while the effect related to MJO waves filtered by the wavenumber–frequency approach is more confined near low latitudes. The average longitude of TC geneses in the MJO–MT category is significantly biased to the west in contrast to those in other categories. This may be a result that the westward-propagating anomaly associated with the MT wave band is likely to develop into a TC after penetrating farther into a MJO-related circulation, which is usually active near the Philippines.

4. Wave characteristics in the different categories

This section focuses on the composite of equatorial waves at the TC genesis time in the categories, in which TC geneses are affected by either a single wave type or multiple wave types. The composite domain has been shifted zonally and meridionally for each case so that the TC centers are coincident with one another. The comparison among the categories is made to reveal the differences in wave characteristics at the time of genesis. They also provide evidence of wave-preferred phase relationships to TC geneses.

Figure 4 shows the composite of wave-filtered anomalies of 850-hPa winds and TRMM rainfall for the MJO, ER, and MT categories. When the MJO wave type alone modulates TC geneses, the mean TC genesis location is situated within the center of MJO-enhanced convection to the north of strong and persistent westerly surge (Fig. 4a). This somewhat differs from the findings by Frank and Roundy (2006). They documented that TC geneses tend to form near the easternmost edge of the low-level westerly surge and within the eastern and equatorward side of a cyclonic gyre by compositing all candidate TCs as long as MJO threshold criterion is met. In this MJO category, because of the lack of the contribution from other wave types, TCs are likely to form in the region of maximum cyclonic vorticity and convection. By comparison, the composite for the ER category shows a classic ER wave pattern in terms of wind and convection anomalies. The positive rainfall anomaly center is located about one quarter wavelength to the east of the cyclonic gyre, and the ER wave propagates more zonally than meridionally, similar to those found in previous studies (e.g., Wheeler et al. 2000; Yang et al. 2007). The robust ER wave signal clearly depicts the phase relationship between wave structure and TC geneses. The mean TC genesis location is situated on the southeastern side of the strong cyclonic gyre (Fig. 4b), consistent with the findings in Frank and Roundy (2006).

For the MT wave band (Fig. 4c), the composited wave exhibits a more zonal propagation, which seems to be distinct from the findings in previous studies (e.g., Lau and Lau 1990; Dickinson and Molinari 2002). They

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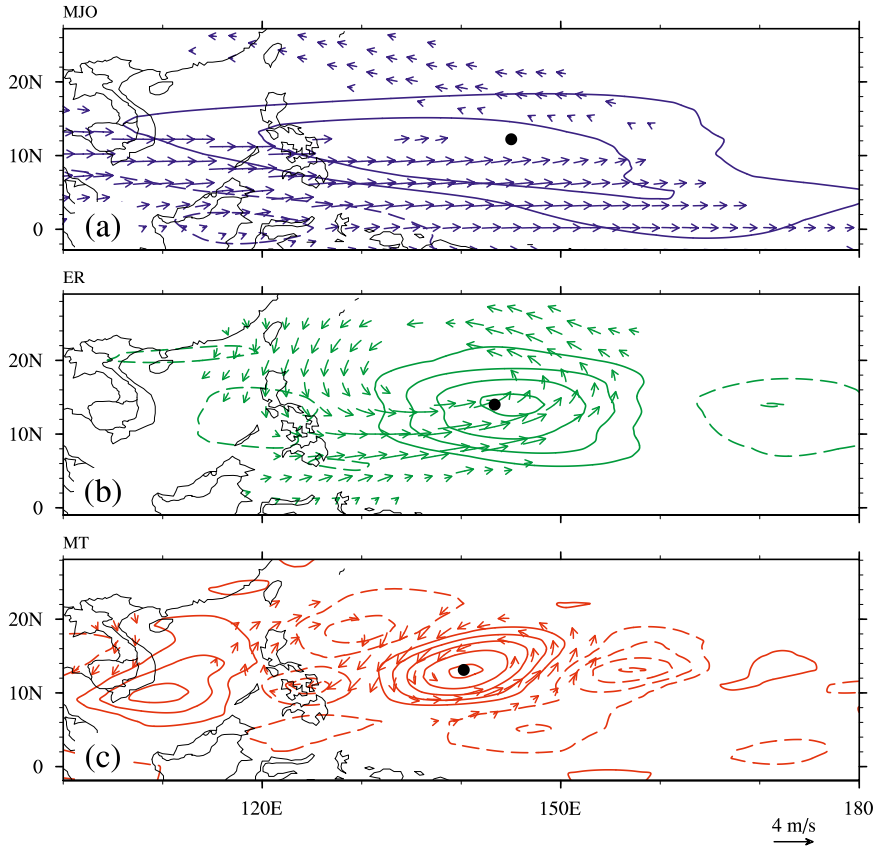


FIG. 4. Composite of the filtered anomalies of 850-hPa winds and TRMM rainfall relative to the TC genesis location and date for the (a) MJO, (b) ER, and (c) MT categories. Rainfall anomalies are contoured at 1 mm day^{-1}. Wind anomalies are plotted when exceeding the 90% confidence level. The black dot stands for the mean TC genesis location in each category.

pointed out that when westward-propagating MRG waves penetrate into a region of a monsoon circulation, they could transit to TD-type disturbances. This transition is accompanied by wave deviation from the equator toward the northwest, a reduction in wavelength and wave amplification, favoring TC geneses. Therefore, a favorable monsoon environment over the WNP plays an important role in modulating synoptic disturbances. In the MT category without the modulation of the environment associated with MJO and ER wave types, the MT wave propagation is found to have a more zonal component at the TC genesis time, although the wave train seems to split into two components around 130°E. The northwestward-propagating branch decays rapidly, and the westward-shifting branch is able to retain the original wave behavior. This wave separation is in agreement with the result in a shallow-water model by Aiyyer and Molinari (2003). In their numerical simulation, when the MRG waves encounter the monsoon trough circulation in response to a heating source, they also split into two parts, one emerging out of the background convergence along the equator and the other growing within the monsoonlike circulation north of the equator (see their Fig. 13). This wave separation could be largely determined by the monsoon circulation over the WNP, such as intensity and vertical structure of the monsoon trough. More observational evidence and modeling examination are needed to further validate this less documented phenomenon in the future.

The composites in the categories involving the effect of multiple wave types take on prominent differences in wave behaviors in contrast to those in the categories only related to a single wave type. For the MJO–ER category, TCs tend to form in the coincident active region of MJO and ER wave bands (Fig. 5a), so that the superimposition of westerly surges associated with these two wave types equatorward of TC can increase the low-level rotation. This indicates that TCs in this category have a tendency to form within an environment with cyclonic horizontal shear of zonal winds, also seen in Fig. 6 for the composite of unfiltered fields for the MJO–ER category. Another distinct feature is that the ER wave displays an elongated

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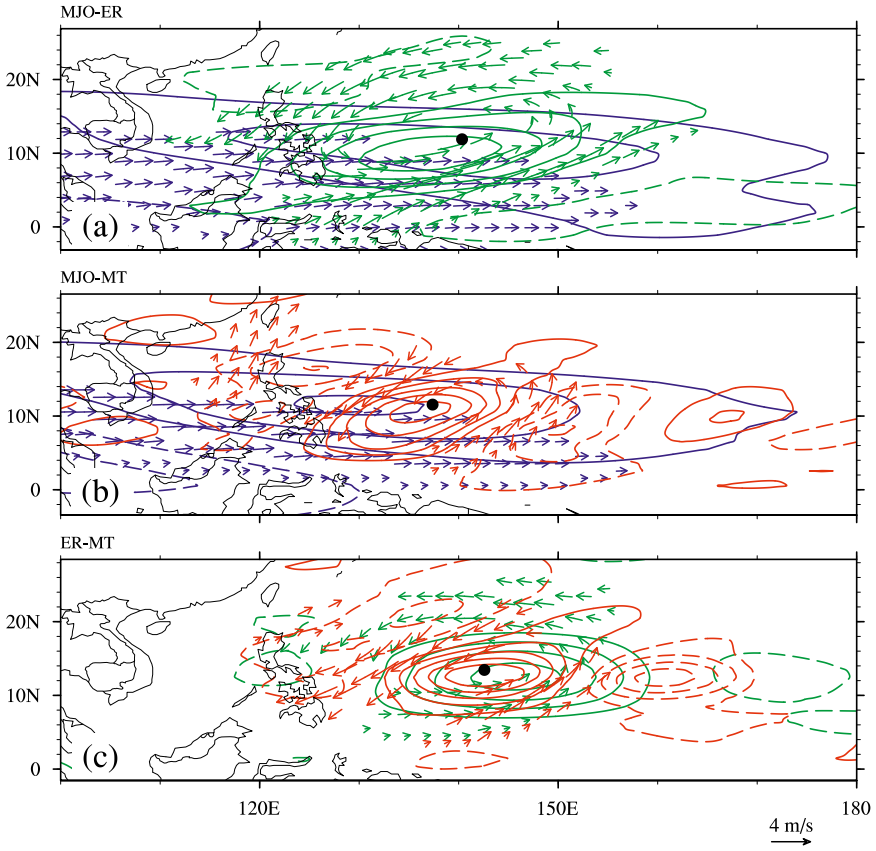


FIG. 5. Composite of the filtered anomalies of 850-hPa winds and TRMM rainfall relative to the TC genesis location and date for the (a) MJO–ER, (b) MJO–MT, and (c) ER–MT categories. The blue, green, and red colors represent the MJO, ER, and MT wave bands, respectively. Rainfall anomalies are contoured at 1 mm day^{-1}. Wind anomalies are plotted when exceeding the 90% confidence level. The black dot stands for the mean TC genesis location in each category.

southwest–northeast-tilted structure and northwestward propagation, which can be identified by a lag composite (not shown). These changed features become much more evident in the categories related to the MT wave as shown in Figs. 5b,c. From a viewpoint of barotropic conversion, the southwest–northeast-tilted structure of ER and MT waves is conducive to produce barotropic conversion from mean kinetic energy to eddy kinetic energy within a monsoonal environment with cyclonic shear, which leads to the stronger wind anomalies of ER (Fig. 5a) and MT (Figs. 5b,c) wave bands compared to the counterparts in the ER and MT categories. This barotropic kinetic energy conversion among the interaction of multiscale systems has been extensively documented in previous studies (e.g., Maloney and Dickinson 2003; Chen 2012).

By comparison with the cases in the MJO category, the TC geneses in the MJO–MT category tend to shift more eastward relative to the MJO-related westerly surge, where the low-level zonal confluence is strong (cf. Figs. 4a and 5b), coincident with the center of cyclonic circulation in the MT wave train. The eastern part of the MT wave train retains the behavior of westward propagation, while the western part deflects northwestward away from the equatorial region when penetrating into the MJO-related monsoon circulation, which differs significantly from that in the MT category in which the wave propagation is dominated by the zonal component along the equator (see Fig. 4c). In addition, the MT wave structure embedded with the MJO-related environment seems to be more coherent and compact than that in the MT category (cf. Figs. 4c and 5b). The disparities in wave characteristics are closely related to the asymmetric monsoon circulation about the equator. Using a shallow-water model, Aiyyer and Molinari (2003) examined the evolution of MRG waves in a monsoon trough environment in response to active MJO convection. The westward-moving equatorially trapped MRG wave could experience scale contraction once entering a region of the

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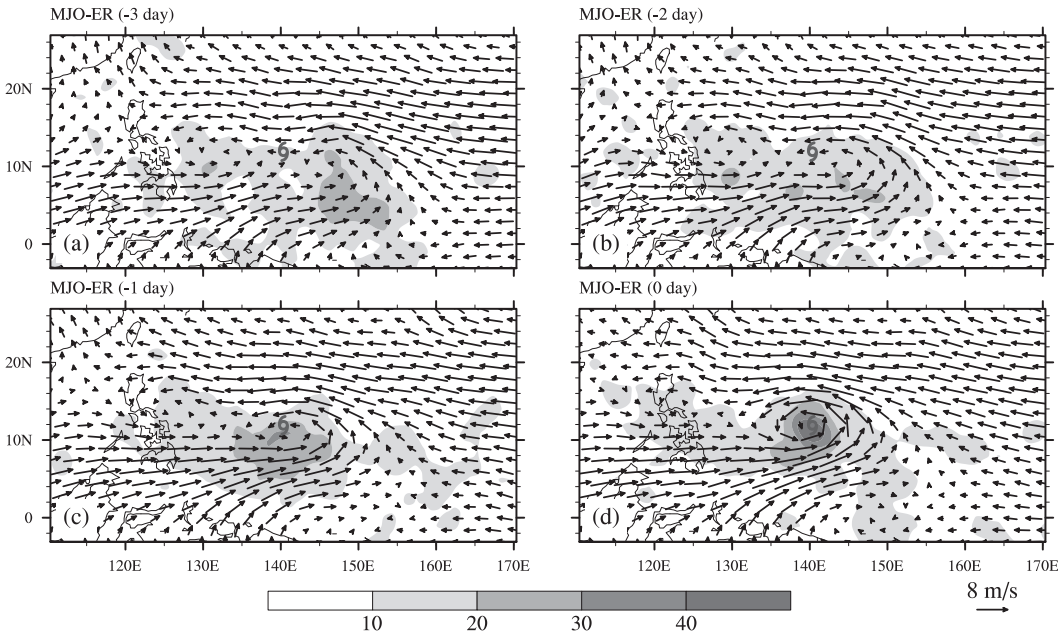


FIG. 6. Composite of total 850-hPa winds and TRMM rainfall relative to the TC genesis location for the MJO–ER category at (a) 3, (b) 2, (c) 1, and (d) 0 days prior to TC genesis. The typhoon symbol represents the mean TC genesis location in the category.

low-level convergence, and then a secondary wave pattern develops along the monsoon trough axis arising from wave–circulation interaction, so that energy is transferred from the basic state to the wave perturbation.

In addition to the modulation of the horizontal fields related to MJO, the MJO-induced westerly surge could alter the intensity of vertical wind shear given a commonly observed first internal mode of the vertical structure of MJO. The vertical wind shear also plays a crucial role in determining the level of wave amplification and the northward propagation of the wave train. Using a 2.5-level model, Wang and Xie (1996) revealed that, in an easterly vertical shear, the baroclinic and barotropic modes are nearly 180° out of phase, whereas they are in phase in the westerly vertical shear. Therefore, an easterly (westerly) vertical shear can lead to the wave amplification at the lower (upper) level. In the MJO–MT category, the increased vertical easterly shear caused by the enhanced westerly surge associated with MJO facilitates to amplify the wave train at the lower-tropospheric level. On the other hand, Jiang et al. (2004) derived a barotropic vorticity tendency equation, in which the vertical wind shear is able to couple barotropic and baroclinic structures. They found that, for a baroclinic structure with a convergence (divergence) in lower (upper) troposphere, a positive (negative) barotropic mode of vorticity could be induced to the north (south) of a cyclonic vortex at lower levels under an environment with an easterly vertical shear. On the basis of the above-mentioned arguments, the vertical easterly shear generated by MJO could favor wave amplification at the lower level and northward displacement of relatively small-scale waves.

Likewise, the MT structure also experiences a similar transition in the ER–MT category (Fig. 5c). TCs tend to form in a region where the ER and MT wave types contribute jointly to enhancing the cyclonic anomaly. Besides, of interest is that the ER-related convection center is seen to be close to the ER-related circulation center in the ER–MT category, so does the case in the MJO–ER category (see Fig. 5a). That is quite different from the counterpart in the ER category without the collaboration with other wave types, in which one-quarter phase difference between vorticity and convection centers is apparent in agreement with the theoretical wave solution in a quiescent background environment (Matsuno 1966). However, in the presence of the cooperation of other wave types, the extent of the coupling of wave circulation and convection could be modified largely, which may be due to the following two processes. On one hand, owing to a broad spatial scale and a long duration, the larger-scale waves, such as the MJO, could alter the background envelope, which modulates the characteristics of smaller-scale waves. Specifically, zonally propagating equatorial waves with loose circulation–convection coupling would gradually deflect from the

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TABLE 1. The zonal wind ($m\,s^{-1}$), zonal convergence of zonal wind ($\times 10^{-6}\,s^{-1}$), and meridional shear of zonal wind ($\times 10^{-6}\,s^{-1}$) averaged in the $4° \times 4°$ domain centered in TC genesis location at 3 days prior to TC genesis in the MJO–ER, MJO–MT, ER–MT, and MJO–ER–MT category. The boldface value in each row represents the most significant environmental feature among the categories.

	MJO–ER	MJO–MT	ER–MT	MJO–ER–MT
u	−1.7	−1.5	−3.0	−1.8
$\partial u/\partial x$	−1.2	−4.0	−2.6	−1.9
$-\partial u/\partial y$	9.4	6.6	2.3	7.7

equator toward the northwest, lose original properties of equatorially trapped waves, and evolve into an organized wave train with tight circulation–convection coupling when penetrating into a favorable asymmetric monsoon circulation over the WNP. On the other hand, smaller-scale waves could in turn exert an upscale feedback to larger-scale waves through the modulation of apparent heat and moisture sources. In other words, the intensity and distribution of nonlinear rectified fields associated with smaller-scale waves could also upscale to affect the extent of circulation–convection coupling of larger-scale waves. More studies are still needed to quantify wave circulation–convection coupling and elucidate how the interaction between multiscale waves impacts the coupling extent.

5. Large-scale environment in the different categories

Some objective methods have been utilized to identify the role of large-scale circulation patterns in TC geneses (e.g., Ritchie and Holland 1999; Yoshida and Ishikawa 2013). By the composites relative to the TC genesis centers, the background circulation can be categorized into different patterns. In the present study, the combination of different equatorial waves related to TC geneses can also partly project onto the unfiltered fields to modulate the large-scale environmental condition during the TC genesis time. This section is devoted to exploring what are the distinct large-scale circulation patterns among the various categories.

The composites of unfiltered wind and TRMM rainfall relative to TC genesis locations can help shed light on the discrepancy of large-scale environment patterns prior to TC geneses for the different categories (Figs. 6–8). For the MJO–ER category, the westerly and easterly flows are prevalent to south and north of the TC genesis location, respectively. As a result, the horizontal shear of zonal winds gives rise to a large-scale monsoon–trough-like circulation pattern several days prior to TC genesis (see Figs. 6a,b and Table 1). Initially, a broad convection region with scattered centers is accompanied by the zonally elongated cyclonic circulation, and subsequently, the scattered convection cells begin to cluster and form an organized cyclonic circulation elliptically encircling the TC genesis location at one day prior to TC genesis. At the time of TC genesis, the cyclonic circulation is tightly contracted in scale and the rainfall is significantly increased around the genesis location (Fig. 6d).

In the MJO–MT category, the monsoon–trough-like shear region is situated to the west of TC genesis three days before TC genesis, which is coincident with a major rainfall region in the vicinity of it (Fig. 7a). This also implies that, at that time, the westerly flows and convection anomalies associated with MJO reside to the west of the TC genesis location. A weak vortex can be discernible to the southeast of TC genesis, which reflects the existence of the MT wave band. The TC genesis region is embedded within an environment dominated by the convergence of westerly and easterly winds (see Table 1), remarkably different from the circulation pattern of cyclonic shear of zonal winds in the MJO–ER category (Fig. 6a). One day later, even though the monsoon trough to the west has no significant migration, the westerly flows to the south of the TC genesis location are enhanced. Moreover, the synoptic vortex coincident with a rainfall center begins to amplify to the southeast of TC, while propagating northwestward toward the TC genesis location (Fig. 7b). By one day before TC genesis, a closed cyclonic circulation forms nearby the TC center, whereas the monsoon trough circulation and rainfall to the west become weakened and retreat westward (Fig. 7c). At the genesis time, the original monsoon trough seems to be obscure and a strong shear trough appears to its east that wraps up a well-organized TC circulation.

In the ER–MT category, the large-scale circulation exhibits a distinct pattern. At several days prior to TC genesis, the TC genesis location is embedded in an environment dominated by easterly flows because of the absence of the MJO-related contribution (Figs. 8a,b). At that time, the rainfall related to westward-propagating ER and MT wave bands has a rather weak signal to the east. One day later, this signal originating from the southeast is intensified and accompanied by a cyclonic vortex forming to the southeast of the TC genesis location (Fig. 8c). Ultimately, a compact vortex centered in the TC genesis location emerges, but with a smaller size compared to those in the MJO–ER and MJO–MT categories (cf. Figs. 6d, 7d, and 8d). This indicates that TC initial size and structure during TC genesis may, to a large degree, depend on precursor waves. During the TC genesis in this category, there is no evident of southwesterly bursts, unlike the cases in the categories involving the MJO wave band. The MJO-related TC geneses tend to

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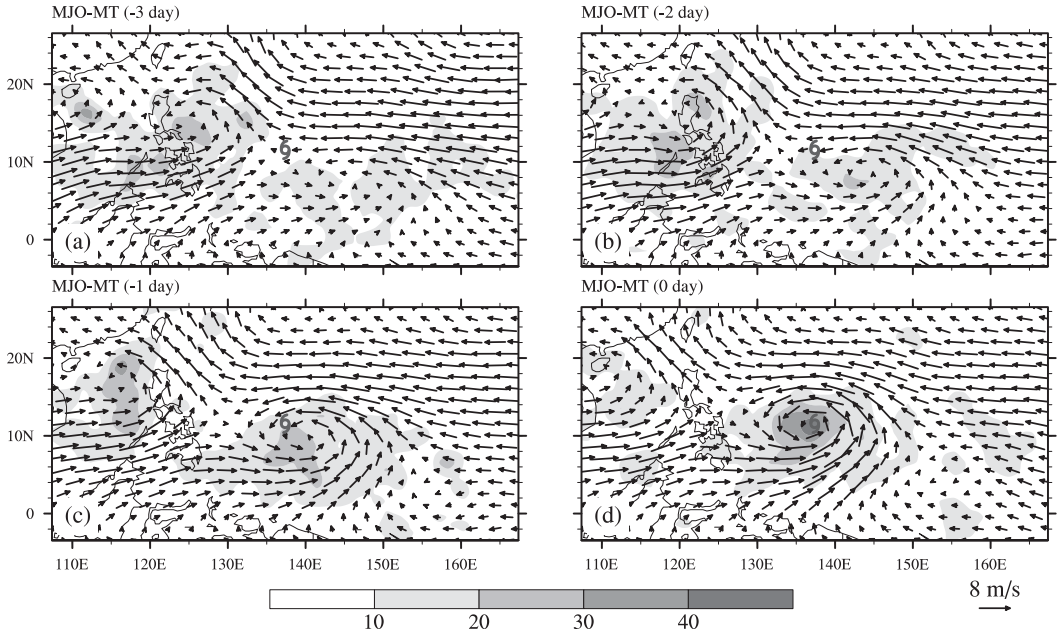


FIG. 7. As in Fig. 6, but for the MJO–MT category.

create a large-sized and relatively incompact structure by external low-level forcing. By comparison, the evolution of TCs with a compact structure in the ER–MT category may be more determined by internal dynamics because of the lack of large-scale environmental forcing.

Regarding the TC geneses related to a single wave type in the MJO, ER, and MT categories, although the large-scale circulation also exhibits some similarities to those in the categories involving two wave types, the precursor signals amplify relatively slowly in the absence of the collaboration of other wave types (not shown), and consequently, smaller-sized and weaker vortices form at the time of TC genesis in contrast to the cases in the categories with more than one wave type. Specifically,

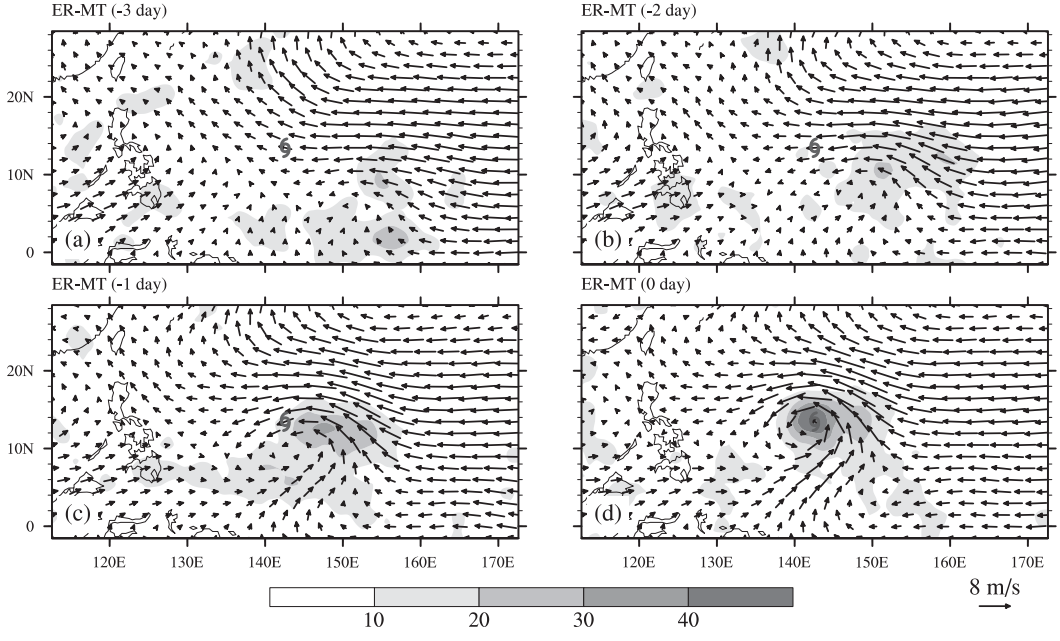


FIG. 8. As in Fig. 6, but for the ER–MT category.

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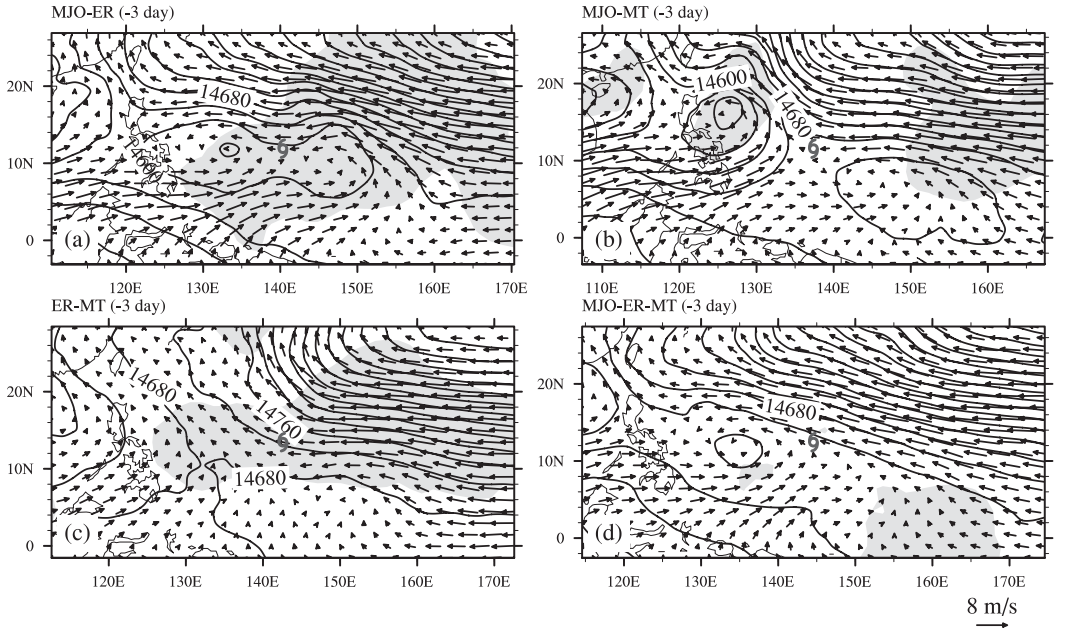


FIG. 9. Composite of the total 850-hPa winds and geopotential height at 3 days prior to TC genesis for the (a) MJO–ER, (b) MJO–MT, (c) ER–MT, and (d) MJO–ER–MT categories. The regions where the geopotential height exceeds the 90% confidence level are shaded. The typhoon symbol represents the mean TC genesis location in each category.

in the MJO category, a monsoon trough is also found to the west of TC genesis as in the MJO–MT category, but has weak intensity in circulation and rainfall fields. Moreover, the westerly wind bursts related to the MJO wave band fail to wrap up a large-scale vortex circulation, due to the absence of ER and MT wave types. This differs from the pattern in the MJO–MT category, in which the external disturbances propagating into the monsoon circulation influence TC geneses. For the cases in the ER category, the TC geneses tend to be embedded within an environment with weak easterly flows prior to genesis. A monsoon–gyre-like circulation then begins to emerge and eventually triggers TC geneses. In the MT category, the disturbances related to the MT wave type seem to be subtle without a pronounced amplification through the interaction with MJO and ER waves. As a result, the TC circulation confined in a small-sized domain does not have strong westerly flows to the south of TC geneses.

The composite of the cases in the MJO–ER–MT category seems to have a hybrid pattern of the large-scale circulations in the MJO–ER, MJO–MT, and ER–MT categories (also see Fig. 9d). The precursor signals incorporate the characteristics of MJO, ER, and MT wave types, leading to a broad cyclonic system with a coherent structure at the genesis time. On the contrary, the TC genesis process in the None category, accounting for a small fraction of TC geneses, does not have significant features associated with the wave bands of interest. The TC geneses may be attributed to other wave bands that are not considered in this study or triggered by upper-level systems.

Ritchie and Holland (1999) showed that a precursor circulation affecting TC geneses appears approximately 72 h before the genesis. To clarify the preferred geographic distribution of TC geneses corresponding to the different categories, Fig. 9 clearly displays the TC genesis locations relative to the monsoon trough and the subtropical high represented by geopotential height 3 days before the genesis. As shown in Fig. 9a, the elongated cyclonic circulation formed by the dominant zonal flows on the southern and northern sides of TC genesis is representative of a well-developed monsoon shear line. Correspondingly, the mean cyclonic shear of zonal wind ($-\partial u/\partial y$) is strongest among the categories in Table 1. This denotes that TC geneses in the MJO–ER category tend to occur in a shear region within monsoon trough, resembling the shear line pattern of TC geneses in the study of Ritchie and Holland (1999). The distinct feature of the MJO–MT category is that the cyclogenesis region is dominated by the zonal wind convergence (Fig. 9b). To the west, the westerly surge establishes a meridionally extending monsoon trough, while, to the east, the easterly flows at the southern edge of the subtropical high are prevalent, producing a strong zonal convergence of winds

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around TC genesis (Table 1). When a synoptic wave train approaches this confluence region, the energy accumulation induced by the confluence of zonal winds can increase TC genesis potential (Zhang and Webster 1989).

In the ER–MT category, TC geneses occur completely within the easterly flows several days before the genesis (Fig. 9c). As seen in Table 1, the mean easterly wind surrounding TC genesis reaches $-3\,\mathrm{m\,s^{-1}}$, stronger than the cases in the other categories. The TC genesis location penetrates more poleward within the easterly environment in the south of the subtropical high, which is responsible for the significantly northward shift of the mean location of TC geneses in the ER–MT category as depicted in Fig. 3. Examination of the circulation composite in the MJO–ER–MT category reveals little evidence for the structure of monsoon trough and the subtropical high (Fig. 9d). To the northeast of the TC genesis location are the uniform southeasterly flows, forming a northwest–southeast orientation of geopotential height. The westerly surge is located to the southwest, establishing a weak vortex to the west of the TC genesis location. The distribution of geopotential height is insignificant, indicating that the large-scale circulation does not exhibit a coherent structure modulating the TC geneses in the MJO–ER–MT category.

6. Conclusions

From the climatological viewpoint of equatorial wave activity, previous studies have examined the impact of different equatorial waves on TC geneses, but did not systematically distinguish the contribution from a single wave or multiple waves. Considering TC geneses over the WNP are closely related to MJO, ER, and MT wave types, the TC geneses are classified into different categories based on the attribution of TC geneses to a single wave or multiple waves. This study makes an attempt to investigate the differences in the wave behaviors and the corresponding large-scale circulations prior to TC geneses in the different categories.

Because TC-projected signals account for a considerable proportion of power spectrum for westward-propagating waves, especially for synoptic waves, the intense rainfall within a TC circulation could largely alter equatorial wave diagnostics (Schreck et al. 2011). To mitigate the contamination, the method of Schreck et al. (2011) was conducted to remove TC-related signals to highlight the impact of major equatorial waves on TC geneses over the WNP. The comparison of the standard deviations contributed from TC geneses for the different wave types exhibits that the synoptic waves are more susceptible to TC contamination in contrast to other wave types with a larger scale. In particular, after removing the TC-related signals, the MT wave type with a spatial and temporal scale close to a TC can reduce the standard deviation by more than 40%, twice as much as the counterpart for the MJO wave type, consistent with the study in Schreck et al. (2011).

Based on the moderate thresholds determined by wave types and TC genesis locations, a total of 214 TC geneses are partitioned into 8 categories, in which the TC geneses are attributed to a single wave type, multiple waves types, or none of wave types. The multiple wave types contribute to about 65% of the TC geneses, which are approximately twice as many as the TC geneses associated with a single wave type. It's worth noting that, without collaboration with other wave types, the MT wave type alone is responsible for less than 10% of the TC geneses comparable to those related to MJO, which seems to differ from the results in Schreck et al. (2012) partly due to different definition of threshold. This suggests that the transition of a MT wave to a TC could become more effective in collaboration with other wave types through their interactive feedback. Among the eight categories, the TC geneses in the ER and ER–MT categories tend to form at high latitudes, significantly different from the cases in other categories. In the absence of MJO-related westerly anomalies, the TC geneses attributed to the ER wave are primarily distributed within the easterly flows in the southern flank of the subtropical high. The TC geneses in the MJO–MT category are more likely to occur farther west, suggesting that the interaction of MJO and MT wave types is conducive to TC geneses in the western part of the WNP.

The composites relative to TC genesis locations demonstrate that the wave characteristics differ remarkably among the categories involving a single wave and multiple waves. In the categories involving a single wave type, the wave types are dominated by the zonally propagating component. In the MJO category, the TC geneses tend to form in a region of maximum positive vorticity to the north of the strong westerly anomalies. In the ER category, the TC geneses are mainly distributed in the eastern part of an ER-filtered cyclonic circulation, due to one-quarter phase lag between the circulation and convection centers for a classical ER wave, which is in good agreement with previous studies (e.g., Fig. 7b in Frank and Roundy 2006). In contrast, the maximum convection coincident with the cyclonic circulation center promotes the TC geneses in the MT category. The combination of more than one wave type can vary wave behaviors and phase relationship with TC genesis locations. The relatively small-scale wave could be enhanced, forming a robust structure. Meanwhile, the propagation of ER and MT waves shifts from westward to northwestward within an environment in association with the MJO wave type. The

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MJO westerly surge can give rise to the low-tropospheric zonal wind convergence, assisting in the energy accumulation of synoptic waves and thus facilitating to cluster TC geneses to the east of the maximum MJO anomalies. The MJO wave could modulate the environmental circulation to enhance the vertical easterly shear, which not only confines the wave amplification at the lower level, but also causes the waves embedded within the MJO-related envelope to propagate northwestward. Therefore, it is observed that the propagation of ER and MT waves has a more northward component in the categories with multiple wave types in contrast to the cases in the categories with a single wave type.

The composites of unfiltered fields relative to TC genesis locations in the multiple-wave categories exhibit distinct large-scale patterns prior to TC geneses. The combination of MJO and ER waves forms a large-scale cyclonic circulation such that the TC geneses in the MJO–ER category are concentrated within a monsoonal shear region to the north of the southwesterly surge. Comparatively, the TCs in the MJO–MT category are more likely to breed in a confluence region to the east of monsoon trough where the zonal wind convergence favors the energy accumulation for the synoptic MT wave. In the ER–MT category, the joint effect of westward-propagating ER and MT wave types cluster TC geneses within the easterly winds in the southern flank of the subtropical high. In contrast, the large-scale circulations related to TC geneses in other categories takes on less significantly distinct patterns. In particular, the composite fields in the MJO–ER–MT category seems to be a complex synthesizing the characteristics of the monsoon shear, monsoon confluence, and easterly flows in the MJO–ER, MJO–MT, and ER–MT category, respectively.

Overall, by the TC genesis categorization in this study, we have clearly shown the differences in wave behaviors and large-scale circulation patterns during the time of TC genesis. The correspondence between multiple waves and large-scale circulation patterns helps us further understand the environmental condition that is favorable for TC geneses. The different combination of multiple wave types could affect the size and intensity of an initial vortex, which to a certain degree provides a hint for TC development at the subsequent time. Application of these results could shed some light on the improvement of operational prediction of TC geneses due to the mutual interaction of multiple equatorial waves.

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